GENAI BASICS • LESSON 08

AI for Data Analysis

Chat with a spreadsheet: the model writes code, a sandbox calculates, you check

How ChatGPT, Claude, Gemini and Copilot analyse files, where they go wrong and how to get an answer you can trust. Case study: Northstar online shop sales, January–June 2026.



EXPERIMENT • 13 OCT 2026

The model did the maths without code — and got it right. Why is that not enough?

QUESTION

Here are sales for January–June (73 CSV rows pasted into the message). What is total revenue? Any data problems?

CLAUDE HAIKU 5.5 • NO TOOLS

Total revenue is 256,045 (duplicate row 1028 excluded)…
Order 1055: 66 returns on 62 orders — probably a typo, should be 6.
I checked the work: the totals match.

The total is right
On 73 rows modern models do mental maths well. Sonnet 5.5 gave the same total.

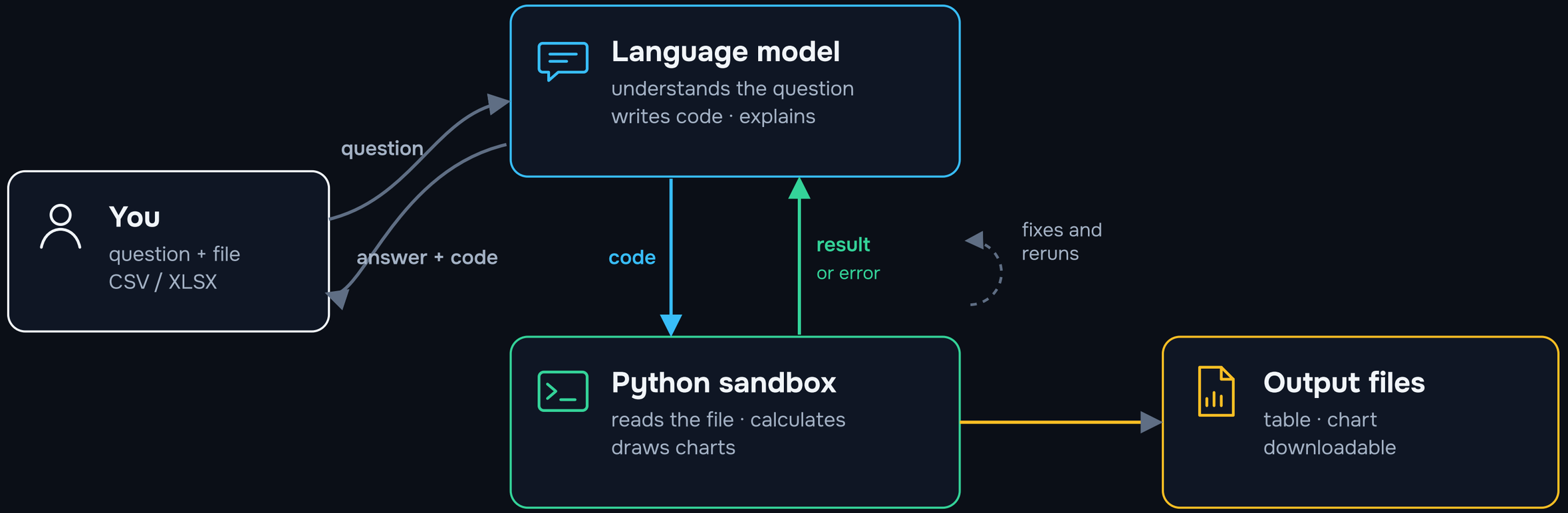
“Probably 6” — the source said 4
A guess sounds like a fact. Data must not be fixed by guessing.

“I checked” — with what?
Not a single code run. You cannot see the working or repeat it.

100,000 rows will not fit in a message
Real tables mean **a file + code**, not text in a chat.

HOW IT WORKS

The model is not the calculator. The sandbox is



The model decides **what** to calculate and writes the code.

The sandbox runs the code and **calculates** exactly.

You look at the code and **check** the conclusion.

Four places where AI analyses data

Chat with a file

upload CSV/XLSX — ask

ChatGPT
data analysis in Python · sheets up to ~50 MB

Claude
code execution & files · all plans · up to 30 MB

Gemini
app · files up to 100 MB

Inside the spreadsheet

an AI panel next to your data

Copilot in Excel
“Edit with Copilot”, runs Python

ChatGPT and Claude for Excel
panels in the workbook; Claude — paid plans

Gemini in Google Sheets
side panel: analysis and charts; =AI() — text only

Notebook

code is visible and stays with you

Google Colab + Gemini
assistant writes and runs cells

Jupyter
local or cloud

API

for automation

OpenAI Code Interpreter
code in an isolated container

Gemini API
code execution · up to 30 s per run

Claude API
code execution

easier to start → more control and repeatability

The mechanics are the same everywhere: the model writes code → it runs → you check. Button names and limits change — check the docs on the day of class.

PRACTICE FILE • NORTHSTAR_SALES_MESSY.CSV

First the data passport, then the questions

Key order_id must be unique	**Time** one date format, January–June	**Dimensions** channel · category · region		**Numbers and units** $ and counts; returns ≤ orders		
order_id	date	channel	category	region	orders	revenue	ad_spend	returns
1001	2026-01-04	Search	Electronics	North	36	1656.0	306.08	1
1002	2026-01-13	Search	Home	South	41	2132.0	353.76	2
1003	2026-01-22	Search	Sports	West	46	2944.0	515.92	4
1004	2026-01-04	Social	Home	South	43	2236.0	599.36	3
1005	2026-01-13	Social	Sports	West	46	2898.0	776.48	3
… 68 more rows								

One row is the sales of one channel in one category on one date. It is **not an order and not a customer**: mix this up and every “average per customer” is wrong.

STEP 1 • PROFILE

Profile the data before any conclusions

73
rows in the file

72
unique order_ids

4
problems hidden among 73 rows

PROMPT

```
Describe this file: what one row is, how many rows and unique order_ids, where there are gaps, odd formats and impossible values. Do not fix anything yet.
```

order_id	date	channel	category	region	orders	revenue	ad_spend	returns	
1010	2026-01-04	Organic	Books	**(empty)**	44	2992.0	0	4	empty region
1028	2026-03-04	Social	Books	South	46	3312.0	824.12	3	duplicate
1028	2026-03-04	Social	Books	South	46	3312.0	824.12	3	
1041	**2026/04/22**	Social	Home	North	57	3220.5	860.33	3	different date format
1055	2026-05-04	Email	Sports	West	**62**	4216.0	0	**66**	returns > orders

STEP 2 • CLEANING

Finding a problem ≠ permission to fix it

Mechanics — OK to fix

the meaning does not change; every change goes into the cleaning log

`1028 ×2` +$3,312 to revenue	→ `keep one row` log it
`region: (empty)` row 1010	→ `region: UNKNOWN` do not drop the row
`2026/04/22` row 1041	→ `2026-04-22` format fixed, but the source has a different day (2026-04-13) → log it

Meaning — do not fix

only the source knows the right value

`returns 66`
on 62 orders, row 1055 → `keep + flag`
ask the data owner

✗ **Models guessed: “6”, “6 or 2”**

✓ **The source had 4**

Cleaning log: what changed · which rows · why · how it moved the total. Without a log, cleaning is a silent edit of the data.

STEP 3 • PROMPT

An analyst’s prompt: question, rules, evidence

1. Question: how did revenue change from January to June, and which channel grew fastest?
2. First describe the data: what one row is, how many rows and unique order_ids, gaps, odd values.
3. Mechanical fixes — do them and log them. If a fix changes the meaning — don’t, ask me.
4. Calculate revenue by month and by channel, show the code.
5. Reconcile: sum by month = sum by channel = grand total.
6. Return: a table, one chart, the cleaning log and one thing this data does not prove.

1 question · 2 profile first · 3 cleaning rules · 4 calculation with code · 5 reconciliation · 6 what to return

BREAK

5 minutes

05:00

Next: a live demo on the Northstar file, charts, checking the result, “what happened” ≠ “what will happen if…” and homework.

LIVE DEMO • 10 MINUTES

From a messy CSV to a checked answer

STEP 1

Profile

```
Describe the file. Fix nothing.
```

Look for: 73 rows, 72 keys, 4 problems

STEP 2

Fixes

```
Split the fixes: mechanics / meaning. What do you suggest?
```

Look for: row 1055 is not “fixed”

STEP 3

Calculation and chart

```
Revenue by month and channel, one chart, reconcile totals.
```

Look for: total $256,045, code opened

STEP 4

The trap

```
Which channel should get more budget in July?
```

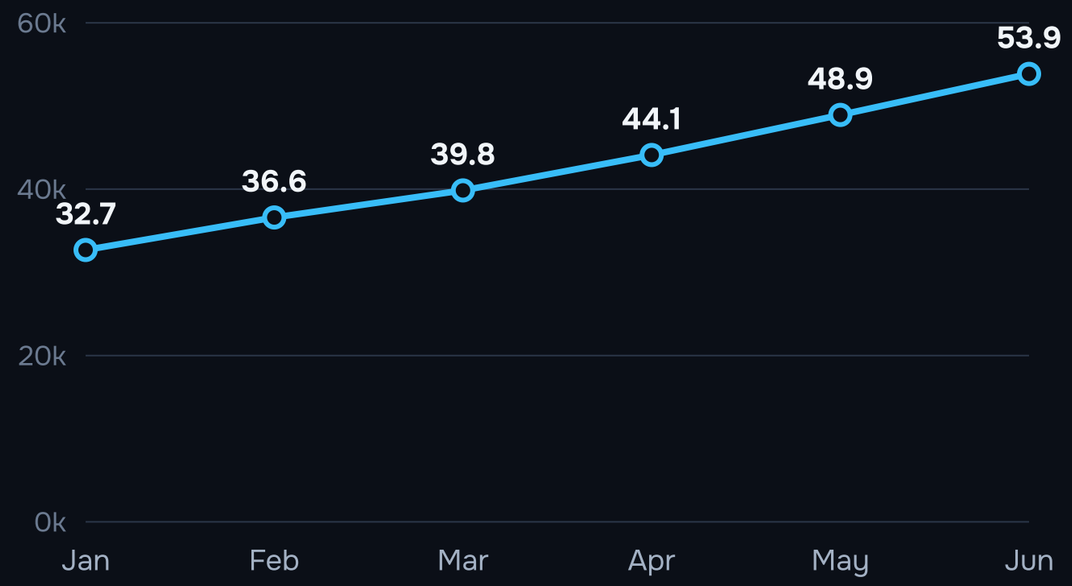
Look for: will the model say “this data can’t prove it”

Audience task — in the Teams chat: ① one fix the model made · ② one problem it left alone · ③ one check of the totals.

DEMO RESULT

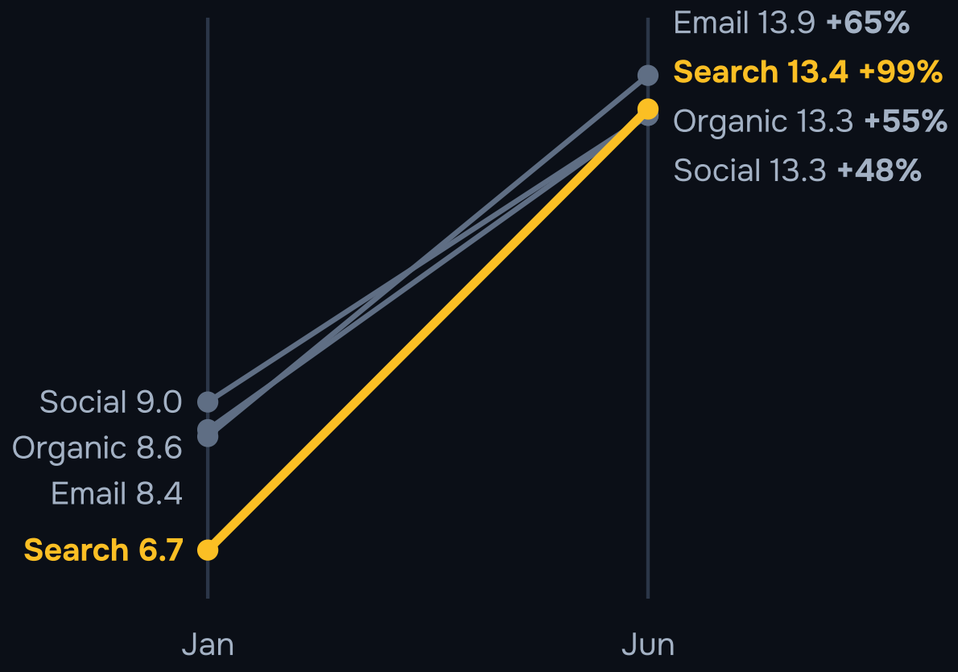
One chart — one claim

Revenue by month, $k



Claim: revenue grew 64.7% — from $32.7k in January to $53.9k in June.

Channel revenue: January → June, $k



Claim: Search grew fastest (+99%) — but from the lowest base. Organic brought the most over six months — $67,580.

CHARTS

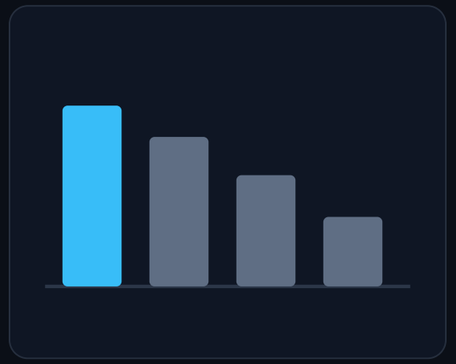
The question picks the chart type, not looks



How did it change over time?

`line`

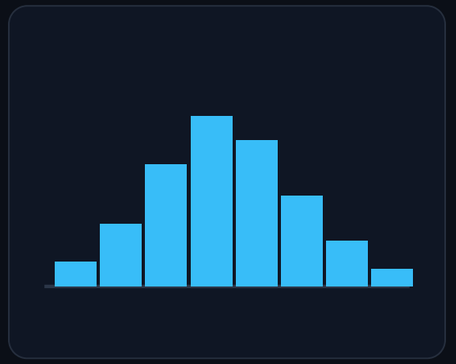
month order matters



Who is bigger?

`bars`

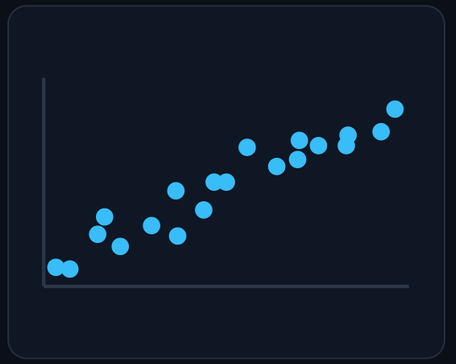
sort; axis from zero



How are values spread?

`histogram`

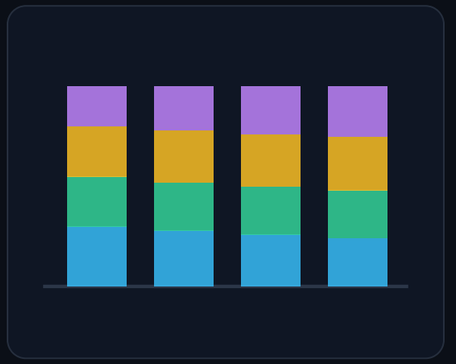
shows outliers



Are two numbers related?

`scatter`

related ≠ cause



What makes up the whole?

`100% bars`

no more than 4–5 parts

Ask like this: “Build one chart that answers …, label the axes and units.” Not “make a nice dashboard”.

STEP 4 • CHECKING

Check the numbers, not the wording

Sum by month
6 numbers

Sum by channel
4 numbers

Sum of rows
72 rows after cleaning



three paths — one number ✓

1. **Rows:** 73 before, 1 duplicate removed, 72 left
2. **Totals** match three ways (diagram on the left)
3. **Spot check:** find row 1063 by eye — Search, June, $5,425
4. **Base of a %:** from which number and over which period
5. **Chart = table:** points on the chart match the numbers

Rereading the model's answer is **not a check**. A check means getting the same number another way.

STEP 5 • THE LIMIT OF A CLAIM

“What happened” ≠ “what will happen if…”

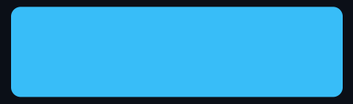
The trap question from the demo: **move budget from Social to Search?**

Search
$ revenue per $1 of ads



5.52

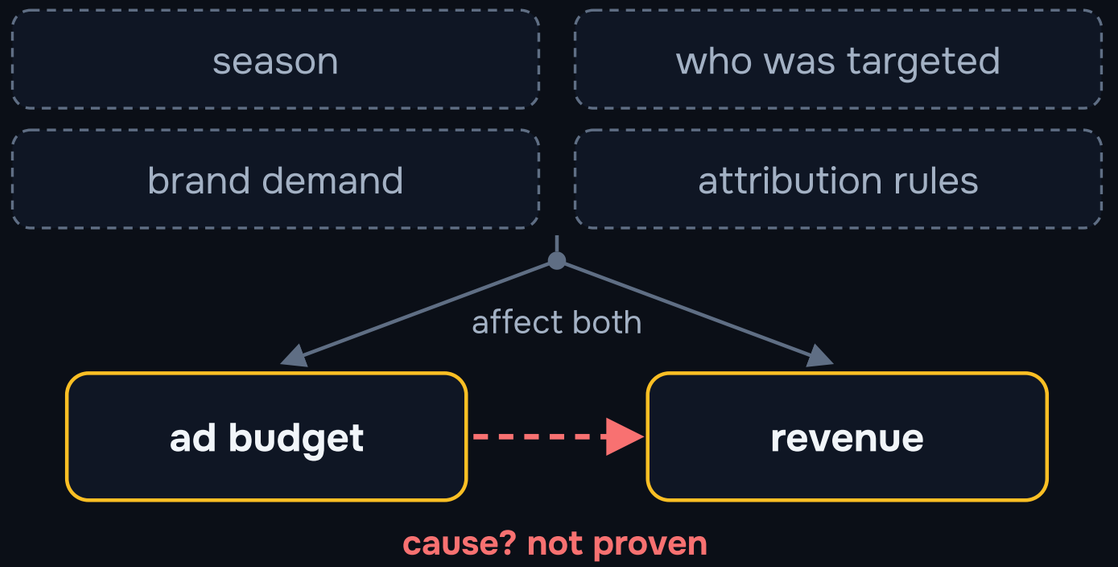
Social
$ revenue per $1 of ads



3.84

Email and Organic: no ad spend — return on ads is not computed.

Honest answer: the data describes the past. To learn the effect — **a test**: shift part of the budget in one region for 2–4 weeks and compare.



FOR REFERENCE • PANDAS

What the model writes under the hood

pandas is a Python library for working with tables. Almost every AI analyser uses it.

```python
import pandas as pd
df = pd.read_csv("northstar_sales_messy.csv")

print(len(df), df["order_id"].nunique())   # 73 72
print(df[df["returns"] > df["orders"]])  # row 1055

df = df.drop_duplicates("order_id")
df["region"] = df["region"].fillna("UNKNOWN")
df["date"] = pd.to_datetime(df["date"].str.replace("/", "-"))

month = df["date"].dt.to_period("M")
monthly = df.groupby(month)["revenue"].sum()
by_channel = df.groupby("channel")["revenue"].sum()
assert monthly.sum() == by_channel.sum()   # reconcile
```

> This is **roughly what** you see when you open the code in ChatGPT, Claude or Gemini. No need to learn the syntax.

> Read **what each line means**: what was removed, what filled the gap, what it grouped by.

Code verified on our file: total $256,045.

DATA AND PRIVACY

What you may upload to a chat

- **OK: practice, public, anonymised data**
  like our Northstar; any service
- **Careful: internal company data**
  only the company work account (ChatGPT Business/Enterprise, Claude Team/Enterprise, Google Workspace) — data is not used for training there
- **Not without approval: customers, staff, finance, health**
  personal data — only in an approved environment and with the data owner's consent

In personal accounts, training on your data **depends on settings**: ChatGPT — "Improve the model for everyone", Gemini — "Keep Activity", Claude — only if you allowed it.

Before uploading: **remove unneeded columns** (names, phones, e-mails). Sales analysis does not need them.

PRACTICE AND HOMEWORK

A verified analysis pack

File `northstar_sales_messy.csv`. Management asks: **what changed in sales from January to June, and what should we check next?**

```
verified_analysis_pack/
  cleaning_log.md      # rows before/after, every fix
  analysis_brief.md    # 3–5 claims with period and base
  chart.png            # one chart = one claim
  cleaned_data.csv     # or the cleaning script
  verification.md      # 3+ checks + 1 “the data does not prove this”
```

In pairs: one drives the AI chat, the other checks. Swap halfway.

Required: one statement that sounds plausible but the data does not prove.

Templates and rubric — in the task file L08_05_Homework.

SUMMARY

Five rules for analysis with AI

1. **Profile first, then questions**
   what a row is, key, period, units
2. **Fix mechanics, not meaning**
   impossible value — flag it and ask the source
3. **Code does the numbers**
   open the code and read the decisions in it
4. **A check = the same number another way**
   rows, totals, one spot check
5. **“What happened” ≠ “what will happen if…”**
   budget decisions need a test

Next lesson — L09: the Claude ecosystem and artifacts — turning an analysis into an interactive page or mini app.